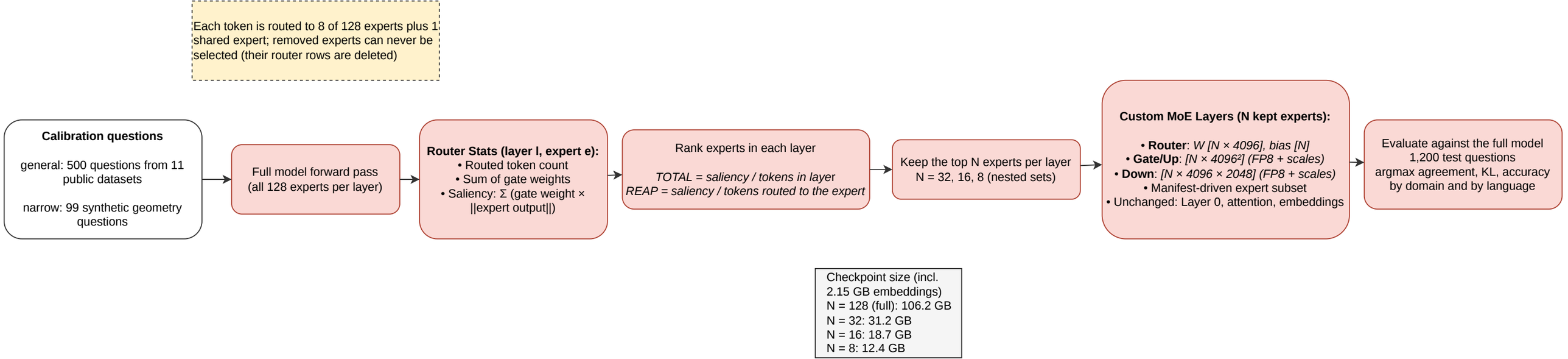
Each token is routed to 8 of 128 experts plus 1 shared expert; removed experts can never be selected (their router rows are deleted)
Calibration questions
general: 500 questions from 11 public datasets
narrow: 99 synthetic geometry questions
Full model forward pass (all 128 experts per layer)
Router Stats (layer l, expert e):
• Routed token count
• Sum of gate weights
• Saliency: Σ (gate weight × ||expert output||)
Rank experts in each layer
TOTAL = saliency / tokens in layer
REAP = saliency / tokens routed to the expert
Keep the top N experts per layer
N = 32, 16, 8 (nested sets)
Custom MoE Layers (N kept experts):
• Router: W [N × 4096], bias [N]
• Gate/Up: [N × 4096²] (FP8 + scales)
• Down: [N × 4096 × 2048] (FP8 + scales)
• Manifest-driven expert subset
• Unchanged: Layer 0, attention, embeddings
Evaluate against the full model
1,200 test questions
argmax agreement, KL, accuracy
by domain and by language
Checkpoint size (incl. 2.15 GB embeddings)
N = 128 (full): 106.2 GB
N = 32: 31.2 GB
N = 16: 18.7 GB
N = 8: 12.4 GB